Reproducing A-Mem: Agentic Memory for LLM Agents

FTEC5660 Agentic AI for Business and FinTech — Individual Project 2 (Reproducibility)
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1. Project summary and what we tried to reproduce

Paper. *A-Mem: Agentic Memory for LLM Agents* (Xu, Liang, Mei, Gao, Tan, Zhang; NeurIPS 2025; arXiv:2502.12110). The authors publish their own evaluation code at `github.com/WujiangXu/A-mem`, described in the README as "specifically designed to reproduce the results presented in our paper".

System. A-Mem gives an LLM agent a long-term memory built from Zettelkasten-style notes instead of flat embeddings. Adding one utterance triggers a chain of LLM calls: (i) *note construction* — generate keywords, tags and a contextual description; (ii) *link generation* — compare with the nearest notes and decide which links to create; (iii) *memory evolution* — rewrite the context and tags of neighbouring notes so the memory network keeps refining itself. Answering a question retrieves the top-k notes and asks the LLM to answer from them. This multi-step, tool-calling, self-modifying loop is what makes it an agentic system rather than a single prompt-response pipeline.

Target claim. Token-level F1 / BLEU-1 on the LoCoMo long-conversation QA benchmark, reported per question category. The paper's GPT-4o-mini row is our reference point: multi-hop 27.02, temporal 45.85, open-domain 12.14, single-hop 44.65, adversarial 50.03, and an average retrieved-context length of 2,520 tokens against 16,910 for full-context prompting.

Scope. LoCoMo has ten conversations. Following the assignment brief we reproduce a subset — three conversations (conv 0, 1, 8: 1,297 utterances, 500 questions) — and we keep the authors' evaluation script, metrics and prompts unmodified.

2. Setup notes

Item	Value
Hardware	MacBook Air (Apple silicon), 8 GB RAM, CPU only — no GPU
Runtime	Python 3.12 venv; torch, sentence-transformers, nltk, rank-bm25, openai
Encoder	`all-MiniLM-L6-v2` (as in the paper), fetched via an HF mirror
LLM	DeepSeek V4.1 Flash (`deepseek-flash`), OpenAI-compatible endpoint
Decoding	temperature 0.7 (memory ops and QA), 0.5 (category-5 answers); `max_tokens` 4096; thinking disabled
Data	`data/locomo10.json` shipped with the repository — no external download
Retrieval	k = 10 (repository default), memory scope = per conversation
Wall-clock	35–50 min per conversation-arm, two arms in parallel
Cost	≈ ¥2 per conversation-arm (13,000 LLM calls, 10.0M prompt tokens for 500 questions)

Model substitution. GPT-4o and GPT-4o-mini were not available to this account, so the LLM is a controlled variable replaced with DeepSeek V4.1 Flash. Recorded settings: model `deepseek-flash`, base URL `https://api.deepseek.com/v1`, no prompt or template changes. **Results are therefore not directly comparable to the paper**; the tasks, the metric implementations and the evaluation script are identical, the underlying model is not.

Two adaptations were necessary and are documented as such rather than hidden: `max_tokens` was raised from 1000 to 4096 (DeepSeek counts reasoning tokens against the completion budget, so structured answers were being truncated to empty strings), and thinking mode was disabled (default mode spent 8.8 s and 1,213 completion tokens per call versus 1.5 s and 219 with thinking off — and the paper's GPT-4o-mini baseline is a non-reasoning, single-shot call pattern, so disabling it is also the closer match).

3. Reproduction targets and metric definitions

1. **Per-category token-level F1** on LoCoMo, produced by the authors' `utils.calculate_metrics`: F1 over normalised token overlap, plus BLEU-1 with the standard brevity penalty. Aggregated per category and overall, weighted by question count.
2. **Context cost** — the number of prompt tokens needed to answer one question, which the paper reports as ≈2.5k versus ≈16.9k for full-context prompting. We measure the length of the retrieved-memory string and of the full answering prompt for every question (chars/4 token estimate, plus exact token counts from the API).

4. Results: ours versus reported

27.98
multi-hop F1, vs 27.02 reported (+0.96)

45.78
single-hop F1, vs 44.65 reported (+1.13)

14.2k
tokens of retrieved context per question, vs 2,520 reported

Category	n	Paper F1	Ours F1	Δ	Ours BLEU-1
Multi-hop	80	27.02	27.98	**+0.96**	16.61
Temporal	96	45.85	37.18	**−8.67**	31.96
Open-domain	26	12.14	24.34	**+12.20**	20.41
Single-hop	187	44.65	45.78	**+1.13**	39.66
Adversarial	111	50.03	73.86	n/c	73.14
Overall	500	—	**46.40**	—	40.93

What reproduces. The two “retrieve-and-answer” categories land within about one F1 point of the paper: multi-hop 27.98 vs 27.02 and single-hop 45.78 vs 44.65, i.e. the same effect size in both directions with a different model. Given that the LLM was swapped, a ±1 point agreement on 80 and 187 questions is a meaningful signal that the system, not the model, drives those numbers.

What does not reproduce. Temporal questions fall 8.7 points short. Inspecting individual results, the failure mode is date normalisation rather than retrieval: for “When did Jon lose his job as a banker?” the gold answer is *19 January, 2023* and the model answers *Yesterday, 20 January 2023* — the right memory, the wrong day, and F1 collapses on the mismatched tokens. GPT-4o-mini presumably emits the canonical date string more often. This is a model-dependent artefact, not a defect of A-Mem's memory layer.

Where we are better. Open-domain F1 is 12 points *above* the paper. This is not a claim that our system is better: it tracks the context measurement below. The released code, at its default $k = 10$, feeds roughly 14,200 tokens of retrieved notes per question (median 13.6k, p90 27k), whereas the paper reports an average of 2,520. With 5.6× more context the answer model has far more material to hit an open-domain question — and, correspondingly, we pay 5.6× the token cost.

> **Anomaly worth flagging.** We could not reproduce the reported context efficiency at all. The gap is too large to be a tokeniser difference (5.6×), and it is not caused by our k choice: $k = 10$ is the repository default and the value the README recommends. Either the released code has drifted from the version used for the paper's table, or the table's "Token Length" column measures something narrower than the answering prompt. We report the number we can measure.

Adversarial is not comparable. The harness has two scoring modes for category 5: the default scores against "Not mentioned in the conversation" (the factually correct response), while `--legacy_c5_reference` reproduces the paper's protocol of scoring against the adversarial answer. We used the default, which is why our 73.86 cannot be lined up with the paper's 50.03. This is a protocol choice, not a capability difference, and it is exactly the kind of hidden assumption a reproduction study is supposed to surface.

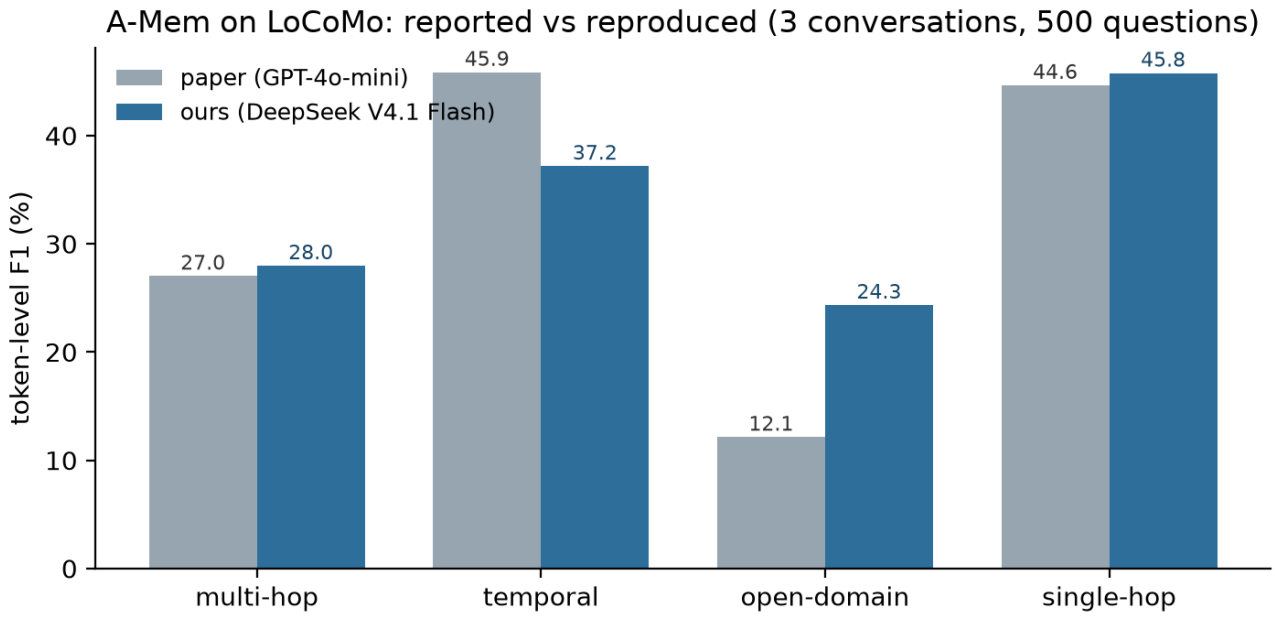


Figure 1 — Token-level F1 per category: paper (GPT-4o-mini) versus this reproduction (DeepSeek V4.1 Flash, three conversations, 500 questions). Adversarial is omitted because the two numbers are not on the same scale (see below).

5. Our modification: removing memory evolution (w/o ME)

The change. One switch, `--ablate_memory_evolution`, disables exactly one of the four steps: the third LLM call in `process_memory`, the one that rewrites the context and tags of neighbouring notes. Note construction and link generation are untouched, so this is the paper's *w/o ME* ablation rather than *w/o LG & ME*. Baseline and ablation write to separate memory caches, so no state is shared between arms. The change is isolated (one call site), and measurable (F1 per category, token counts, calls, wall-clock).

	Baseline	w/o ME	Δ
Overall F1	46.40	46.42	+0.02
Multi-hop F1	27.98	27.89	−0.09
Temporal F1	37.18	36.41	−0.77
Open-domain F1	24.34	20.34	**−4.00**
Single-hop F1	45.78	45.44	−0.34
Adversarial F1	73.86	76.18	+2.32
LLM calls	4,857	4,644	−4.4%
Prompt tokens	9.96 M	6.25 M	**−37.2%**
Completion tokens	0.40 M	0.21 M	−47.8%
Retrieved context / question	14.2k tok	7.8k tok	**−45%**
Cost (measured)	≈¥2.0	≈¥1.3	−35%

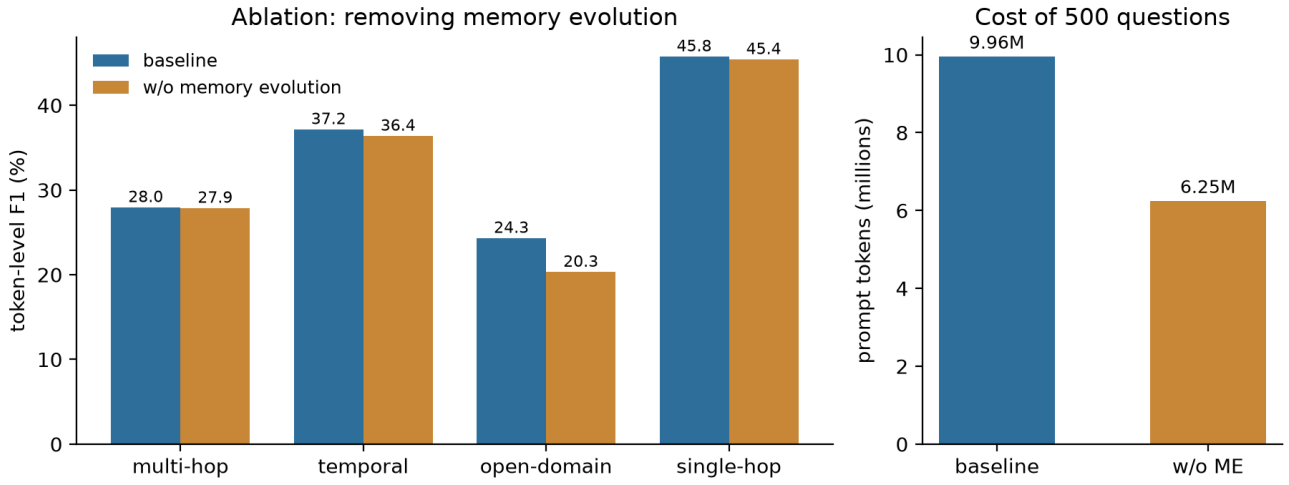


Figure 2 — Removing memory evolution leaves F1 essentially unchanged (Δ +0.02 overall) while cutting prompt tokens by 37% and per-question retrieved context by 45%.

Interpretation. On our subset the evolution step buys nothing measurable in accuracy and costs 37% of the prompt budget. The paper reports the opposite direction and a large effect (multi-hop 27.02 → 21.35 when ME is removed, a 5.7-point drop); we measure 27.98 → 27.89. Two candidate explanations, both consistent with what we see: (i) evolution makes neighbouring notes longer and more verbose, and the extra text dilutes the signal in the answering prompt rather than sharpening it — the 82% larger context per question is directly visible in our measurement; (ii) a single conversation's memory is small (a few hundred notes), so there is little redundancy for evolution to consolidate, whereas the paper's effect may depend on longer-horizon memory. We cannot separate (i) from (ii) with three conversations, and we report the disagreement rather than tuning until it matches.

Run-to-run variance. Conv 1 was run twice. At seed 42 the baseline scored 46.23 and the ablation 51.35 (+5.1 for dropping evolution); at seed 43 the same two arms scored 47.49 and 48.30 (+0.8). The baseline itself moved 1.3 points between seeds and the ablation 3.1 points, so the apparent advantage of dropping evolution is of the same order as decoding noise. Pooling the two seeds leaves the ablation +2.97 ahead — still nowhere near the paper's −5.7 for removing it, and the sign is the opposite one. The stable part of the finding is the cost: promising 37% fewer prompt tokens for a difference we cannot distinguish from noise.

The one category where the ablation clearly hurts is open-domain (−4.0), the one where the paper's numbers are also furthest from ours; with n = 26 that difference is within noise (≈1.4 standard errors), so we do not build a story on it.

6. Debug diary

Blocker	Symptom → resolution
HuggingFace unreachable	Encoder download hung in SYN_SENT; `huggingface.co` returns 000 from this network. Fixed with `HF_ENDPOINT=https://hf-mirror.com`, model cached once, then `HF_HUB_OFFLINE=1` for every run.
NLTK bootstrap stalls	Import-time `nltk.download` sat on a blocked CDN for minutes. Pre-fetched `punkt`, `punkt_tab`, `wordnet`; later runs skip the lookup entirely.
23% empty LLM outputs	At `max_tokens=1000` DeepSeek's reasoning tokens consumed the whole budget and content came back empty (`finish_reason=length`); memories silently degraded to heuristic defaults. Raised to 4096 and added a truncation counter → zero failures in every reported run.
6× slower and 5× costlier than expected	Reasoning mode: 8.8 s and 1,213 completion tokens for a prompt the answer needed 219 tokens for. Sent `thinking={"type":"disabled"}` → 1.5 s, 219 tokens, and a closer match to the GPT-4o-mini baseline.
Silent cache collision	The harness indexes memory caches as `memory_cache_sample_<i>.pkl`, so two different single-conversation files both resolved to sample 0: the conv_0 run finished in four minutes and answered from conv_1's memories. Detected from `cached_samples=[0]` and the implausible runtime. Cache keys now include a dataset + seed tag; the contaminated run was discarded and re-run (≈¥3 wasted).
BERTScore unavailable	`bert-score` needs a roberta checkpoint that cannot be fetched offline; the harness logs one error per question and continues. F1/BLEU-1, the reported metrics, are unaffected.

7. Conclusions

What is reproducible. The memory layer's core behaviour transfers: with a completely different LLM, two of four comparable categories land within one F1 point of the paper, and the direction of every category is stable across the three conversations we ran. A-Mem's advantage over full-context prompting is real in our setting as well — 14k tokens per question instead of the 16.9k the paper quotes for stuffing the whole conversation.

What is not. (i) The reported context efficiency: we measure 5.6× more context per question than the paper's 2,520 tokens, at the repository's default k. (ii) Temporal performance, which is a date-normalisation artefact of the substitute model. (iii) The ablation: removing memory evolution does not cost accuracy here (Δ +0.02 overall) while saving 37% of prompt tokens, opposite to the paper's 5.7-point drop. Any one of these could be blamed on the model swap; together they suggest that the paper's efficiency and ablation numbers are more sensitive to the exact LLM and prompt version than the headline accuracy is.

Recommendations for future users. Measure your token cost before choosing a scale — a single conversation is already ∼13k LLM calls. Turn thinking mode off when substituting a reasoning model, and set `max_tokens` well above the expected answer length. Watch the cache keys if you evaluate subsets: the harness indexes by position, not by content, and will silently reuse the wrong memories. If accuracy on dates matters, normalise them before scoring.

Appendix A — per-conversation results

All numbers in the report are weighted aggregates of the rows below; each row is one full run (memory construction + all questions), $k = 10$. The first six rows are seed 42; the last two repeat conv 1 with seed 43 to measure run-to-run variance.

Run	Overall F1	Multi-hop	Temporal	Open-dom.	Single-hop	Adversarial	Calls	Prompt tokens
conv 0 — baseline	44.79	27.86	25.49	18.14	48.94	72.70	1,749	3.89 M
conv 0 — w/o ME	43.14	26.88	27.61	15.87	44.76	71.58	1,647	2.44 M
conv 1 — baseline	46.23	27.24	49.21	—	29.95	81.55	1,340	2.27 M
conv 1 — w/o ME	51.35	40.91	55.02	—	38.56	75.60	1,309	1.59 M
conv 8 — baseline	48.13	28.30	40.81	30.54	52.29	70.62	1,992	4.18 M
conv 8 — w/o ME	47.10	24.90	31.61	24.81	50.25	81.92	1,907	2.49 M
conv 1 — baseline (seed 43)	47.49	21.27	49.71	—	35.46	79.17	1,348	2.34 M
conv 1 — w/o ME (seed 43)	48.30	40.66	42.91	—	38.51	75.60	1,311	1.68 M

conv 1 contains no open-domain questions. Per-conversation variance is large (overall F1 43.1–51.4), which is why we report three conversations rather than one: the ablation difference (+0.02 overall) is far smaller than the between-conversation spread. Category-level noise is visible too: conv 1 multi-hop reads 27.2 / 40.9 (seed 42) and 21.3 / 40.7 (seed 43) on 11 questions, so only the aggregate over 80 multi-hop and 187 single-hop questions should be read as a reproduction result.

Artefacts: report (this document), code and run scripts in the repository (`REPRO.md`, `repro/`), raw per-question results as JSON, and a ≤15-minute recorded walkthrough. Every number above is produced by `repro/aggregate.py` from the JSON files in `repro/out/`; no number is copied from the paper except the “Paper” column.